

P_{100} is the continuous (100% ED) excitation power at which the coil attains temperature T_{max} with the part mounted to a massive heatsink at 20°C

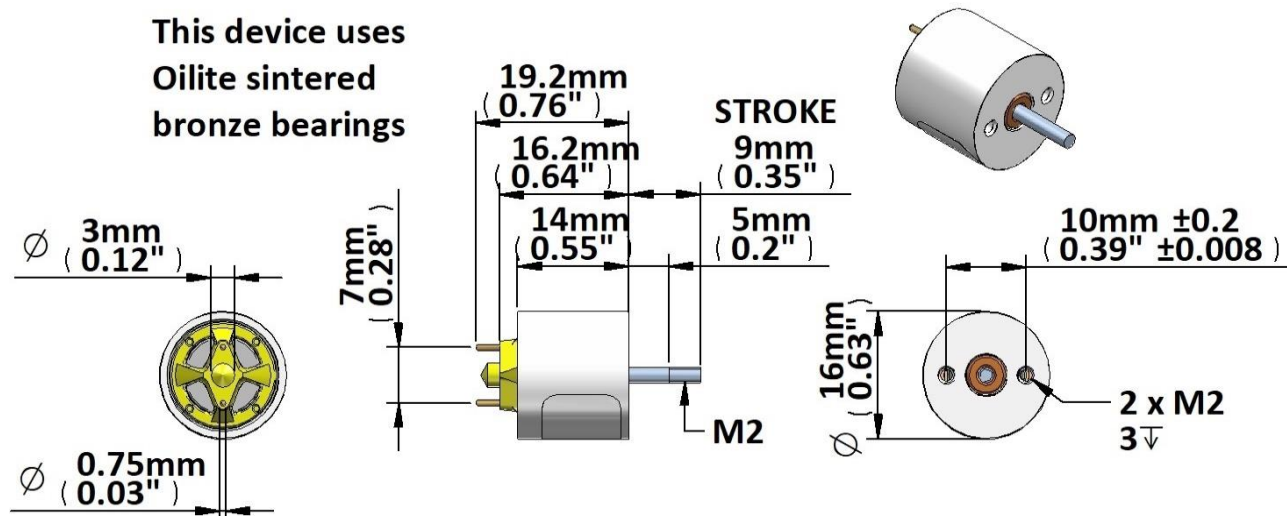
P_{100} 5 W
 T_{max} 130 °C

Total Mass 15 g

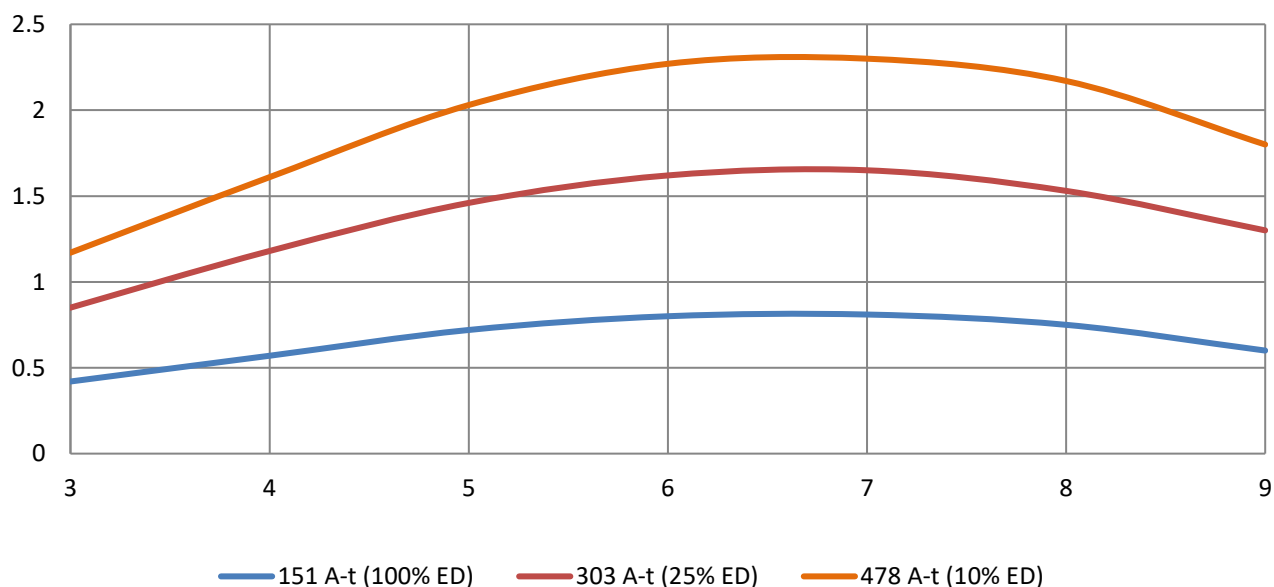
Coil Mass 3 g

Model No.	Resistance R_{20}	Inductance	Force Constant	Velocity Constant	Current I_{100}
VM1614-200	2.6 Ω	0.2 mH	0.7 N/A	0.7 Vs/m	1172 mA
VM1614-180	3.5 Ω	0.3 mH	0.7 N/A	0.7 Vs/m	1010 mA
VM1614-125	15.0 Ω	0.8 mH	1.5 N/A	1.5 Vs/m	488 mA
VM1614-100	39.0 Ω	4.0 mH	2.4 N/A	2.4 Vs/m	303 mA

	Max 'ON' time	Peak Force
100% ED	∞	0.8 N
50% ED	22 s	1.1 N
25% ED	9 s	1.7 N
10% ED	3 s	2.3 N



Force (N) vs Displacement (mm)



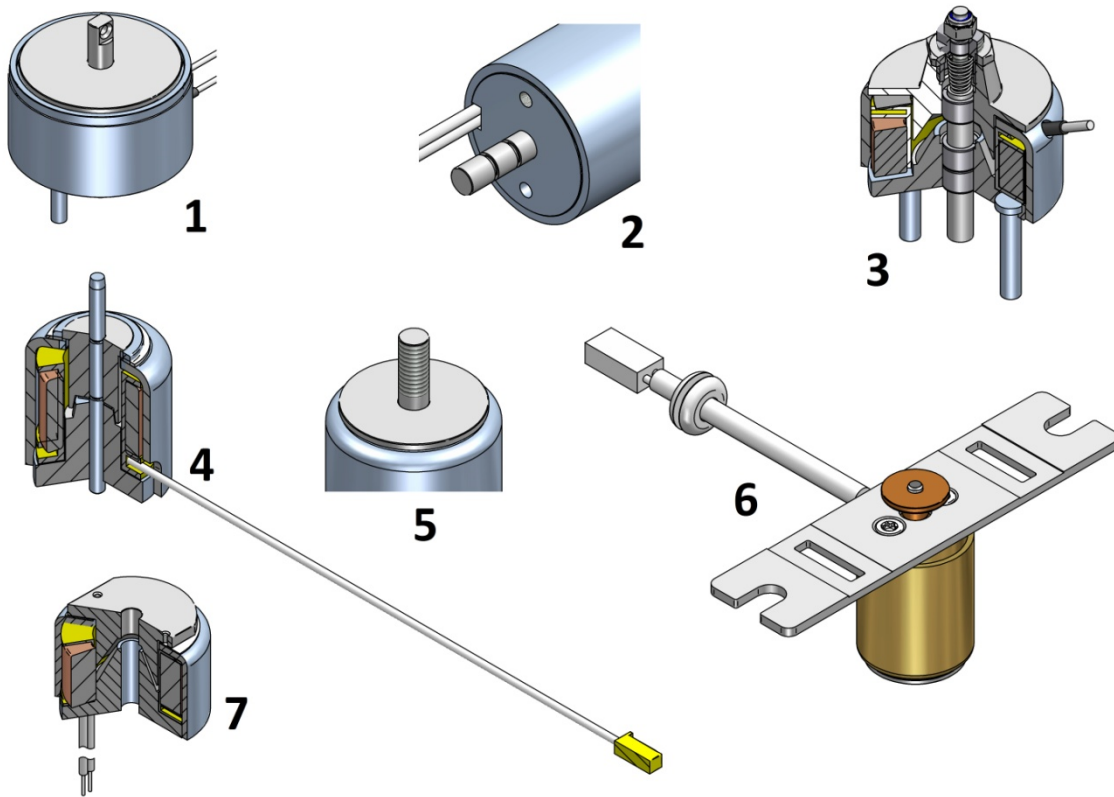


Selection Process for Push-Pull Solenoid

1. Metric (M prefix) and SAE (F prefix) screw thread options are available.
2. The solenoid size is determined based on required force, displacement, and duty cycle from force-stroke characteristic graphs in the solenoid datasheets. Note that this may also be influenced by available power and speed requirements, for a given force requirement a larger solenoid will develop the required force with lower power input, however the higher moving mass may make this slower in operation than a smaller device
3. The pole piece form is also selected from the characteristic graphs, some sizes are available with either flat or conical polepiece design as standard options (note that intermediate or other force characteristic may be possible with polepiece geometry customisation)
4. The coil requirements are determined from tables of coil gauge / duty cycle for the chosen size of device. Coil rating is specified as AWG size of the coil wire
5. The life expectancy of the solenoid is specified by the suffix, P is standard life (2M-5M cycles), PE is extended (5M-10M cycles). For the small push-pull solenoids a different bearing construction is used with special heat-treatment of the bore for nominal >5M cycles. Life expectancy is very much application specific, it will be reduced by long stroke, excessive side loading, particulate contamination and corrosive or otherwise aggressive environments. But may be increased by short stroke, low side loading and clean operating conditions. With the right environment and application setting it is possible to achieve Life expectancy of +50Million cycles. As life expectancy is application specific it needs to be verified under real operating conditions in the customer application to ensure this is sufficient for purpose.

Customisation of the Push-Pull Solenoid

Most of the attachment components of the push-pull solenoid are produced by machining and are amenable to modification even in small (100's or less) quantity. Some typical examples are illustrated below.



1. Flats and cross-hole machined in shaft at armature side
2. Grooves machined in shaft at base side
3. Shaft decoupled from plunger by spring, maintenance-free bearings
4. Modified plunger with shallow angle for increased force at extended position, shaft hardened with sphere end on base side tapered on armature side, and lead wire assembly with connector
5. Screw threads machined on shaft on armature side
6. Mounting plate, bronze bush pressed on shaft, custom lead assembly
7. Modified armature with flat sides and threaded holes, no shaft



GEEPLUS Push Pull Solenoid size 110

Device drawn in energised condition

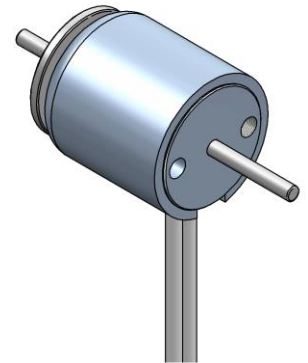
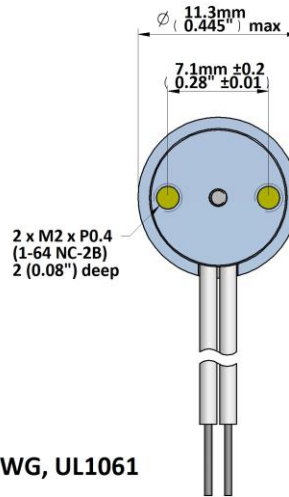
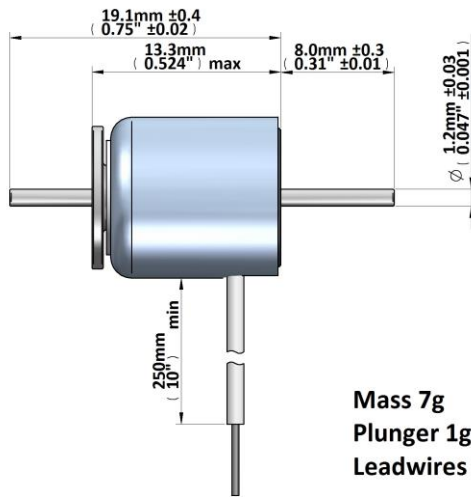
conical plunger

Life Expectancy (cycles): >5M

Available mechanical options:

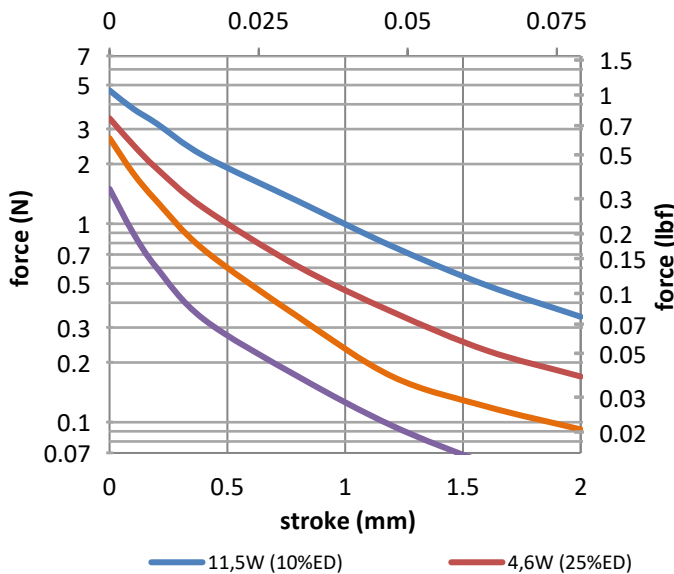
M: metric thread

F: SAE thread

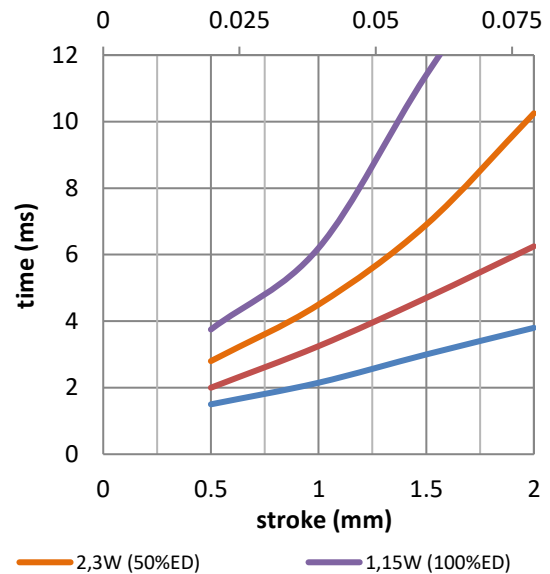


Mass 7g
Plunger 1g
Leadwires 28AWG, UL1061

force / stroke 110C
stroke (inch)



response time
stroke (inch)



Data at 20°C, without heatsink

duty cycle = $\frac{\text{"on" time}}{\text{"on" time} + \text{"off" time}} \times 100\%$			100% cont.	50% or less	25% or less	10% or less
Max. "on" time in seconds			∞	100	36	7
watts at 20°C			1.15	2.3	4.6	11.5
ampere-turns at 20°			105	148	210	332
type no.	resistance $\Omega \pm 10\%$ (at 20°C)	number of turns	volts DC			
M110C-3V F110C-3V	10.5	390	3.0	4.2	6.0	9.5
M110C-6V F110C-6V	31.5	700	6.0	8.5	12	19
M110C-12V F110C-12V	143.0	1450	12	17	24	38

Insulation Resistance >100MΩ, 500VDC Megger
Class A (105°C) insulation class

Dielectric Strength 1000VAC, 50/60Hz, 1 minute

Geeplus reserves the right to change specifications without notice

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GEEPLUS Push Pull Solenoid size 141

Device drawn in energised condition

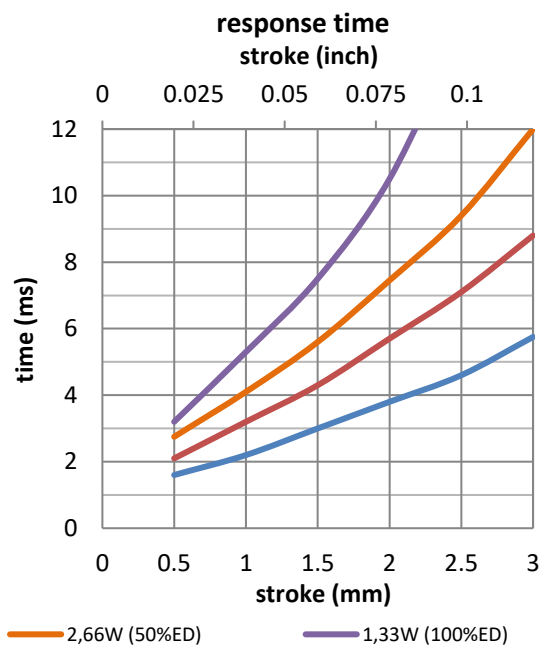
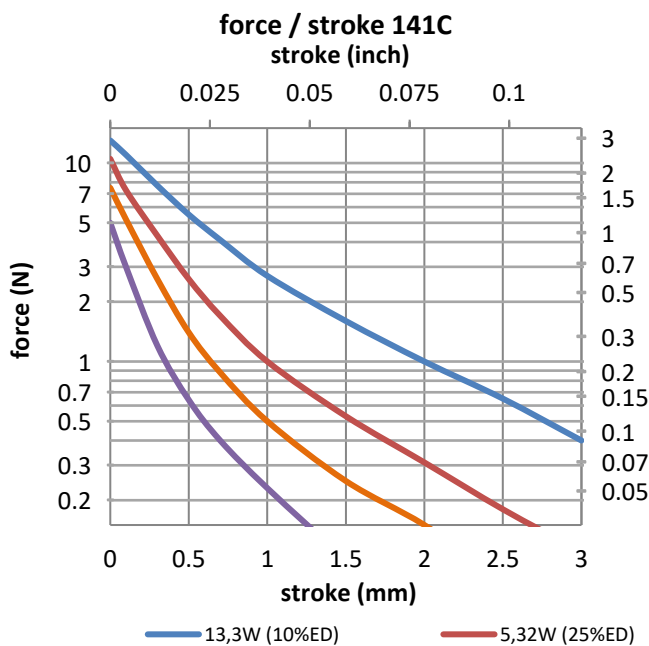
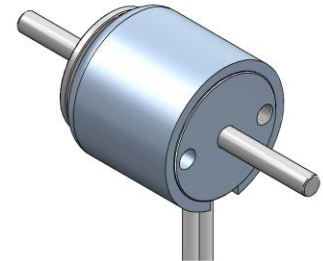
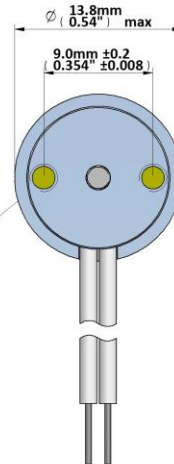
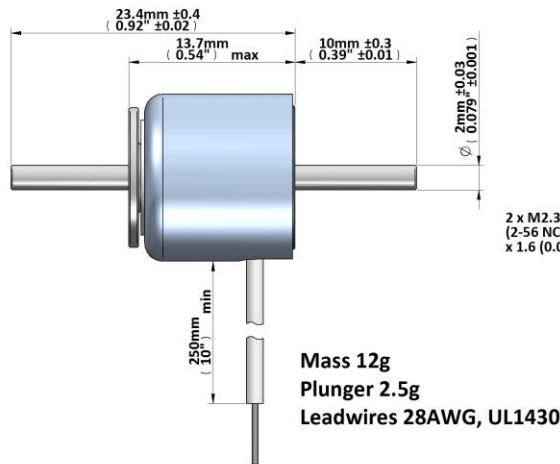
conical plunger

Life Expectancy (cycles): >5M

Available mechanical options:

M: metric thread

F: SAE thread



Data at 20°C, without heatsink

duty cycle = $\frac{\text{"on" time}}{\text{"on" time} + \text{"off" time}} \times 100\%$			100% cont.	50% or less	25% or less	10% or less
Max. "on" time in seconds			∞	100	36	7
watts at 20°C			1.33	2.66	5.32	13.3
ampere-turns at 20°			133	189	267	422
type no.	resistance	number of turns	volts DC			
	$\Omega \pm 10\%$ (at 20°C)					
M141C-3V F141C-3V	6.5	330	3.0	4.2	6.0	9.5
M141C-6V F141C-6V	30	700	6.0	8.5	12	19
M141C-12V F141C-12V	97	1200	12	17	24	38
M141C-24V F141C-24V	468	2600	24	34	48	76

Insulation Resistance >100MΩ, 500VDC Megger

Dielectric Strength 1000VAC, 50/60Hz, 1 minute

Class E (120°C) insulation class

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GEEPLUS Push Pull Solenoid size 144

Device drawn in energised condition

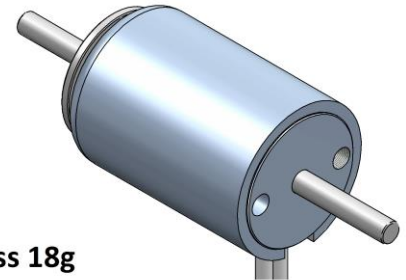
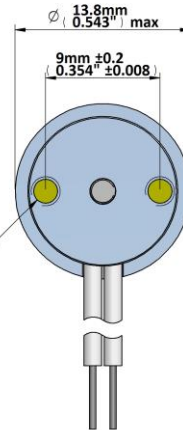
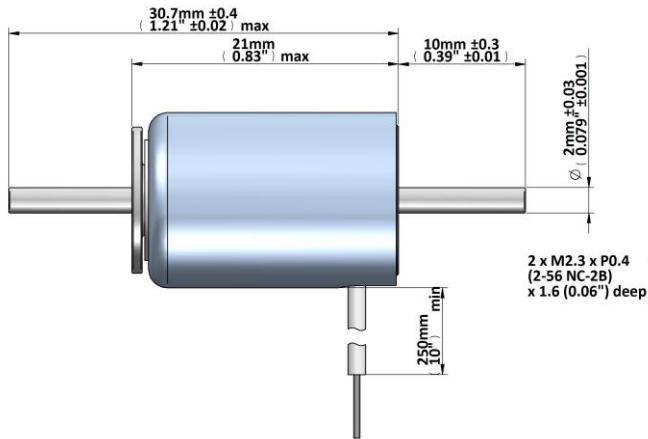
conical plunger

Life Expectancy (cycles): >5M

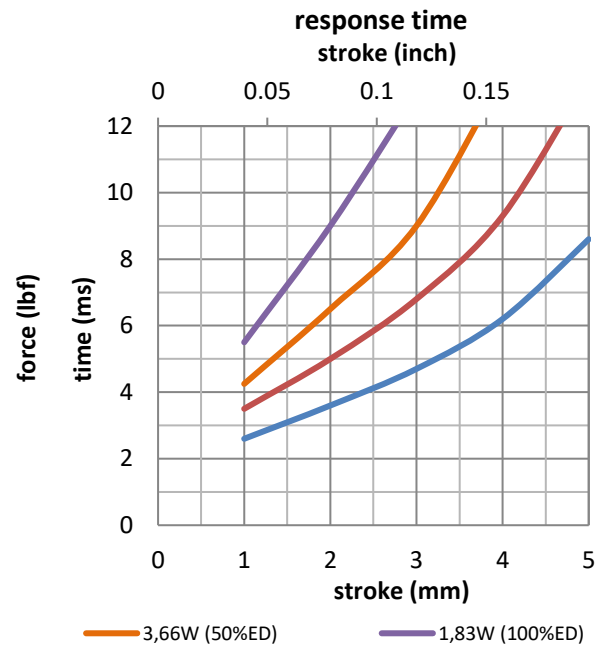
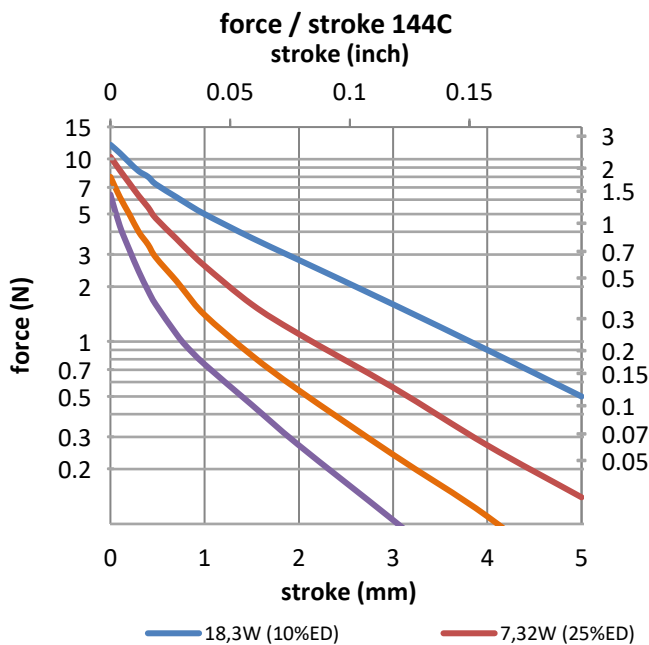
Available mechanical options:

M: metric thread

F: SAE thread



Mass 18g
Plunger 3g
Leadwires 28AWG,
UL1430



Data at 20°C, without heatsink

duty cycle = $\frac{\text{"on" time}}{\text{"on" time} + \text{"off" time}} \times 100\%$			100% cont.	50% or less	25% or less	10% or less
Max. "on" time in seconds			∞	100	36	7
watts at 20°C			1.83	3.66	7.32	18.3
ampere-turns at 20°			236	334	472	746
type no.	resistance $\Omega \pm 10\%$ (at 20°C)	number of turns	volts DC			
M144C-3V F144C-3V	5.0	415	3.0	4.2	6.0	9.5
M144C-6V F144C-6V	22.7	910	6.0	8.5	12	19
M144C-12V F144C-12V	91.5	1750	12	17	24	38
M144C-24V F144C-24V	329	3150	24	34	48	76

Insulation Resistance >100M Ω , 500VDC Megger

Dielectric Strength 1000VAC, 50/60Hz, 1 minute

Class E (120°C) insulation class

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